

PSP APG Minor Version Upgrade(Manually)

- Minor Version Upgrade

If you enable auto minor version upgrades on your DB instance, minor version upgrades occur automatically.

You need to ensure that the Application Team is aware of the Scheduled Maintenance Window of your Minor Version Upgrade, since AWS sometimes doesn't notify priorly for the DB Minor Version Upgrade.

Reference AWS Docs:https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_UpgradeDBInstance.Updating.html#USER_UpgradeDBInstance.Updating.Manual

Regional Cluster Upgrade:

Pre-upgrade:

Steps 1: stop application and take a snapshot of your database before upgrading.

Step 2: connect Database and check aurora version. also check any pending maintenance activity.

```
[postgres=> SELECT aurora_version();
 aurora_version
-----
 14.3.1
(1 row)
```

Step 3: check connections

```
SELECT dbname as db,username as
  username,client_addr as machine, count(*) FROM pg_stat_activity
where username not in ('rdsadmin','postgres')
group by db,username,machine;
```

Upgarde from AWS Console:

Step 4: connect AWS console and select APG cluster

select **Modify**

RDS > Databases > spsp-sys-db

spsp-sys-db

ModifyActions

Related

Filter by databases

	DB identifier	Role	Engine	Region & AZ	Size	Status	CPU	Current activity
	spsp-sys-db	Regional cluster	Aurora PostgreSQL	us-west-2	2 instances	Available	-	
	spsp-sys-db2	Writer instance	Aurora PostgreSQL	us-west-2c	db.r6g.2xlarge	Available	2.52%	0.00 sessions
	spsp-sys-db1	Reader instance	Aurora PostgreSQL	us-west-2b	db.r6g.2xlarge	Available	1.98%	0.00 sessions

Step 5: change DB engine version 14.3 to 14.5

Modify DB cluster: spsp-sys-db

Settings

DB engine version

Version number of the database engine to be used for this database

Aurora PostgreSQL (Compatible with PostgreSQL 14.3) ▼

DB cluster identifier [Info](#)

Enter a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region.

spsp-sys-db

The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Step 6: click **continue** , Click on **Apply Immediately** . will take sometime to upgrade the cluster.

Note: if you are planning to upgrade the database instantly, or else Click the other option to upgrade during the next scheduled Maintenance Window and Modify DB Instance.

Upgrade Via IPS:

Change parameters in Config.json and prereq.json

```
"engineMajorVersion" : "14",
"engineMinorVersion" : "3",
```

to

```
"engineMajorVersion" : "14",
"engineMinorVersion" : "5",
```

Deploy:

```
cdk list --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_pre-req.json
cdk list --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_config.json
```

```
cdk diff --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_pre-req.json
cdk diff --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_config.json
```

```
cdk deploy --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_pre-req.json
cdk deploy --all -c env=pds -c group=primary-pds-mon -c config=psp_mon_config.json
```

Post-Upgrade:

Step 7: once upgraded , please check engine version and other parameters, Instance versions

```
postgres=> select aurora_version();
 aurora_version
-----
14.5.1
(1 row)
```

Connectivity & security	Monitoring	Logs & events	Configuration	Maintenance	Tags
Instance					
Configuration		Instance class		Storage	
DB instance ID spip-sys-db2		Instance class db.r6g.2xlarge		Encryption Enabled	
Engine version 14.5		vCPU 8		AWS KMS key spip-sys-db-cmk 🔗	
DB name sysapgb		RAM 64 GB		Storage type -	
Option groups default:aurora-postgresql-14 🔗 In sync		Availability		Performance Insights	
Amazon Resource Name (ARN) arn:aws:rds:us-west-2:152430470825:db:spip-sys-db2		Failover priority 1		Performance Insights enabled Turned on	
Resource ID db-XXB5U56HOO2BWIMY544KJBCN44				AWS KMS key aws/rds 🔗	
Created time September 11, 2022, 09:09 (UTC+05:30)				Retention period 7 days	
Parameter group spip-shared-sysbobpg14reader65c42ce5-ycwrhcqdxson 🔗 In sync					

Step 6: To list your currently installed extensions.(for major version upgrade need to upgrade)

```
SELECT * FROM pg_extension;
```

To update an extension after an engine upgrade,

```
ALTER EXTENSION extension_name UPDATE TO 'new_version';
```

Step 7: check connections

```
SELECT datname as db,username as
      username,client_addr as machine, count(*) FROM pg_stat_activity
where username not in ('rdsadmin','postgres')
group by db,username,machine;
```

Global Database Upgrade Minor Version(14.3 to 14.5):

For a minor upgrade on an Aurora global database, you upgrade all of the secondary clusters before you upgrade the primary cluster.

AWS Link: <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-global-database-upgrade.html#aurora-global-database-upgrade.minor>

Pre-Upgrade:

Step 1: Please connect Primary and secondary cluster and check aurora Postgres version

Primary Cluster:

```
[postgres=> SELECT aurora_version();
 aurora_version
-----
 14.3.1
(1 row)
```

Secondary Cluster:

```
(postgres=> SELECT aurora_version();
 aurora_version
-----
 14.3.3
(1 row)
```

Apply patch and make it into same version.

Step 2: check any pending maintenance fro both clusters

Maintenance

Maintenance window
mon:07:00-mon:07:30 UTC (GMT)

Pending maintenance
available

Pending maintenance (1)

Filter by pending maintenance

Description	Type	Status	Apply date
Upgrade to Aurora PostgreSQL 14.3.5	system-update	available	-

it will have downtime. apply patch on both primary and secondary clusters and make it into same version.

Step 3: Apply patch on both primary and secondary clusters and make it into consistent.

Upgrade Via AWS console:

Step 4: First , we can upgrade secondary cluster 14.3 to 14.5.(same like above Regional cluster upgrade)

Step 5: Next , we can upgrade secondary cluster 14.3 to 14.5.(same like above Regional cluster upgrade).

Step 6: we don' t need to upgrade global Database , it will happen automatically.

Upgrade via IPS:

Upgrade Via IPS:

Change parameters in Config.json and prepeq.json

```
"engineMajorVersion" : "14",
"engineMinorVersion" : "3",
```

to

```
"engineMajorVersion" : "14",
"engineMinorVersion" : "5",
```

Deploy for secondary cluster and Primary cluster:

```
cdk list --all -c env=pds -c group=primary-pds -c config=psp-prerequisites-pds.json
cdk list --all -c env=pds -c group=primary-pds -c config=psp_pds_ibob.json
```

```
cdk diff --all -c env=pds -c group=primary-pds -c config=psp-prerequisites-pds.json
cdk diff --all -c env=pds -c group=primary-pds -c config=psp_pds_ibob.json
```

```
cdk deploy --all -c env=pds -c group=primary-pds -c config=psp-prerequisites-pds.json
cdk deploy --all -c env=pds -c group=primary-pds -c config=psp_pds_ibob.json
```

Post-Upgrade:

Step 7: To list your currently installed extensions.(not mandatory for minor version)

```
SELECT * FROM pg_extension;
```

To update an extension after an engine upgrade

```
ALTER EXTENSION extension_name UPDATE TO 'new_version';
```

Step 7: check connections

```
SELECT datname as db,username as  
       username,client_addr as machine, count(*) FROM pg_stat_activity  
where username not in ('rdsadmin','postgres')  
group by db,username,machine;
```

NOTE : Once the database is upgraded, it wont be possible to revert back the changes. If you need the Older version of your database, you need to manually restore it from the snapshot. The DB Snapshot is taken before and after the upgrade activity.